

What the program must do

REQUIREMENTS

1. Show one question at a time. Three in total.
2. The student types a short answer.
3. Say right or wrong after each answer.
4. Keep a score: right, asked, percent.
5. "New Question" moves to the next one.
6. After question 3, show END OF QUIZ.
7. Refuse bad input. Do not count it.
8. "Exit" closes the program.

THE THREE QUESTIONS

Can a VB project have more than one form?

answer: Y

A label is a control that permits user input.

answer: T

The '/' symbol is used for comments in VB.

answer: F

The handout picture shows "Question #5". It is from an old version. We do 3.

One form, four Subs, six variables

VARIABLES — OUTSIDE EVERY SUB

```
Public Class Form1

    Private Question1, Question2,
        Question3 As String
    Private QuestionNumber, QuestionAsked,
        QuestionRight As Integer
    Private PercentRight As Decimal
```

Inside a Sub, a variable dies at End Sub.

Your score would reset to 0 on every click.

So declare them here, at the top of the class.

FOUR SUBS — NOTHING ELSE RUNS

Form1_Load

Set up. Show question 1.

Button1_Click

Check Answer: grade, count, lock.

Button2_Click

New Question: clear, move, show.

Button3_Click

Exit.

Load → Check → New → Check → New → Check → end

Set everything up, then show Q1

WHAT IT DOES

Runs one time, when the form opens.

Saves the three questions into variables.

Sets the counters to 0.

Shows question 1.

Hides both result labels.

Turns "New Question" off.

YOU NEED

& joins text CStr() number to text

.Visible = False .Enabled = False

```
Question1 = "Can a VB project have ..."  
Question2 = "T/F: A label is a ..."  
Question3 = "T/F: The '/' symbol is ..."  
  
QuestionNumber = 1  
QuestionAsked  = 0  
QuestionRight  = 0  
  
Label1.Text = "Question #" & CStr(QuestionNumber)  
TextBox1.Text = Question1  
  
Label3.Visible = False  
Label4.Visible = False  
Button2.Enabled = False
```

purple = the lines students forget

Read the answer, refuse bad input

WHAT IT DOES

Reads the answer box.

Cleans it: no spaces, all upper case.

Stops if the box is empty.

Stops if the letter is not Y, N, T or F.

Exit Sub means: leave now, count nothing.

YOU NEED

Trim() removes spaces

UCase() turns y into Y

MsgBox() shows a message

Select Case ... Case Else

```
Dim CurrentAnswer As String

CurrentAnswer = UCase(Trim(TextBox2.Text))

If CurrentAnswer = "" Then
    MsgBox("Please type an answer first.")
    Exit Sub
End If

Select Case CurrentAnswer
    Case "Y", "N", "T", "F"
        ' good answer, keep going
    Case Else
        MsgBox("Use one letter: Y, N, T or F.")
        Exit Sub
End Select
```

Grade it, count it, then lock it

WHAT IT DOES

Select Case picks the right answer for this question. If correct, add 1 to QuestionRight. Then update the score box.

Last step: turn Check Answer OFF and New Question ON. Without this, one question can be counted many times.

1. GRADE IT

```
Label3.Visible = True

Select Case QuestionNumber
    Case 1
        If CurrentAnswer = "Y" Then
            QuestionRight = QuestionRight + 1
            Label3.Text = "You are right!!"
        Else
            Label3.Text = "WRONG."
        End If
    Case 2    ' same, but "T"
    Case 3    ' same, but "F"
End Select
```

2. COUNT IT, THEN LOCK IT

```
QuestionAsked = QuestionAsked + 1
PercentRight = QuestionRight / QuestionAsked
TextBox3.Text = "Questions Right: " & ...

Button1.Enabled = False

If QuestionNumber < 3 Then
    Button2.Enabled = True
Else
    Button2.Enabled = False
    Label4.Visible = True
    Label4.Text = "END OF QUIZ."
End If
```

Clear, move forward, show

WHAT IT DOES

Clears the answer box.

Hides the old result label.

Adds 1 to QuestionNumber.

Select Case shows that question.

Gives the turn back to Check Answer.

It never grades anything.

YOU NEED

Select Case .Visible = False

.Enabled = True / False

```
TextBox2.Text = ""
Label3.Visible = False

QuestionNumber = QuestionNumber + 1
Label1.Text = "Question #" & CStr(QuestionNumber)

Select Case QuestionNumber
    Case 1
        TextBox1.Text = Question1
    Case 2
        TextBox1.Text = Question2
    Case 3
        TextBox1.Text = Question3
End Select

Button1.Enabled = True
Button2.Enabled = False
```

Exit

EXIT

```
Private Sub Button3_Click(...)  
    End  
End Sub
```

One line. That is the whole module.

(Me.Close() is the cleaner way, if you know it.)